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Sivasankar Natarajan  · 3rd+
Technical Director | GenAI Practitioner | Azure Cloud Architect [...]
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Most enterprises are trying to build advanced AI agents without mastering the basics.
That's why 80% of agent projects never make it past the pilot stage.

Here's the 3-layer progression that actually works:

BASIC LAYER (Foundation)

1. Large Language Models (LLMs)
 - Models that generate human-like text and answers from enterprise prompts and data
 - Get this right first—everything builds on model selection and deployment
2. Prompt Engineering
 - Designing structured prompts so models respond consistently, safely, and in the required format
 - 80% of reliability issues stem from prompt quality, not model capability
3. APIs & External Data Access
 - Connecting AI to internal tools and SaaS via secure APIs, SDKs, and webhooks
 - Without data access, your LLM is just an expensive chatbot
4. RAG for Knowledge Bases
 - Retrieval-Augmented Generation: grounding LLM answers in trusted enterprise data
 - This is where generic AI becomes domain-specific AI

INTERMEDIATE LAYER (Capability)

5. Context Management
 - Handling long conversations, session history, and workflow state across steps, channels, and users
 - Stateless agents can't handle real enterprise workflows
6. Memory & Retrieval Mechanisms
 - Short-term and long-term memory so agents can "learn" from past events, runs, and feedback
 - Without memory, every interaction starts from zero
7. Function Calling & Tool Use
 - Allowing agents to call tools, scripts, and APIs to take real actions—not just answer text
 - The leap from chatbot to agent happens here
8. Multi-Step Reasoning

- Breaking complex goals into smaller subtasks with planning, reflection, and verification
- Simple queries need one step; enterprise workflows need orchestrated sequences

9. Agent-Oriented Frameworks

- Frameworks for orchestrating multi-agent systems, tools, and workflows in production
- This is where you move from "one agent doing one thing" to "agent systems"

ADVANCED LAYER (Autonomy)

10. Agentic Workflows

- End-to-end workflows where specialized agents collaborate across Dev, Sec, and Ops
- Multiple agents working together, each handling their domain

11. Autonomous Planning & Decision-Making

- Agents that set sub-goals, pick tools, and adapt plans based on real-time signals and constraints
- Static workflows become dynamic strategies

12. Self-Learning & Feedback Loops

- Continuous improvement using user feedback, evaluations, run metrics, and A/B tests
- Agents that get better over time without manual intervention

13. Fully Autonomous Cloud-Scale Agents

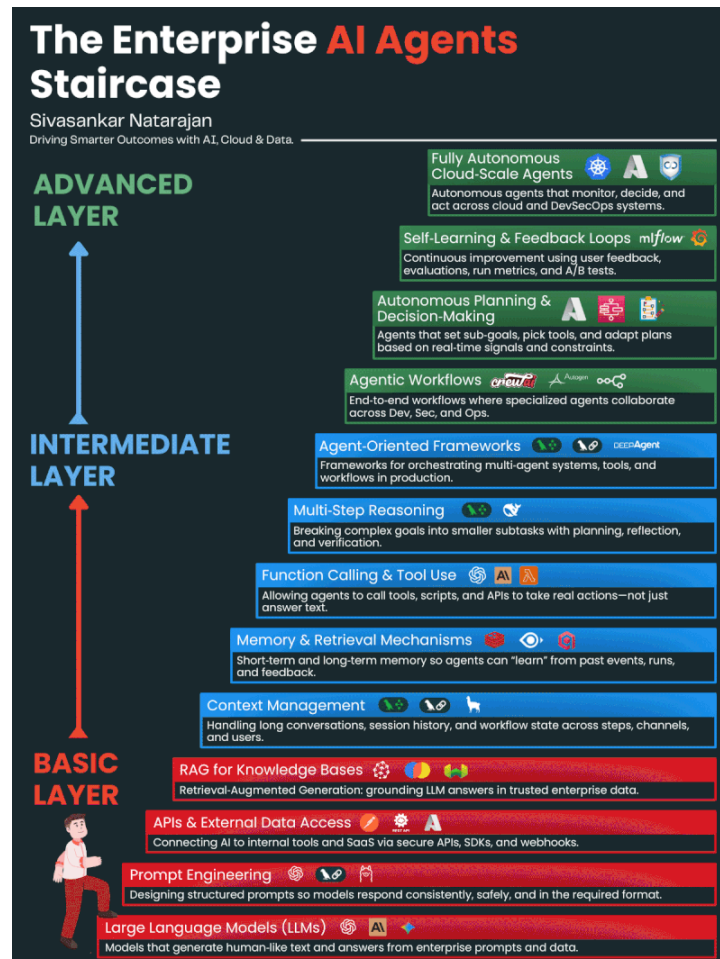
- Autonomous agents that monitor, decide, and act across cloud and DevSecOps systems
- The destination: agents operating independently at enterprise scale

Which layer is your team actually at? And which layer do you think you're at?

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Himani Bansal · 3rd+

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Google Certified Data Analyst | Analytics Consultant, turning data into confi...

This is a great breakdown. So many teams jump straight to “agentic” solutions without solid data access, prompts, and context handling. Foundations decide whether anything scales beyond a pilot.

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Sivasankar Natarajan · Author

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Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Himani Bansal Exactly. Pilots are easy, but scaling requires that technical rigor

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Harish Agoram · 3rd+

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AI Transformation Leader | Agentic AI Strategy for Enterprise Scale Producti...

This is the right ladder...and most teams overestimate their position by one full layer. Until context, memory, and tool reliability are boring and observable, anything labeled “autonomous” is just a fragile demo with a bigger prompt.

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Sivasankar Natarajan · Author

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Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Harish Agoram Exactly. A big prompt is just a brittle substitute for a proper architecture.

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Adrian Maliszewski · 3rd+

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AI Safety & Systems Engineer | Deterministic, Fail-Closed AI Systems | For...

Great discussion topic — thank you for sharing this perspective on GenAI and agentic applications in enterprise contexts.
From a systems and safety engineering standpoint, one core challenge is not only what these agent workflows can automate but how the system-level invariants and safety constraints are enforced across autonomous trar ...more

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Sivasankar Natarajan · Author

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Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Adrian Maliszewski Great point. We see success using monitoring layers to audit state changes..

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AUROBINDA MONDAL · 3rd+

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AI & Cloud Engineering Leader | Career Freedom Coach for Senior Tech Pro...

Many teams aim straight for advanced AI agents but skipping the basics leads to failed pilots. Start with strong models, solid prompts and reliable data before layering memory, reasoning and autonomy.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

AUROBINDA MONDAL Exactly. A pilot without a foundation is just a demo that won't scale..

Like | Reply

 **Sonali Kurade**  · 3rd+ 2d ...
Co-founder | DevOps Consultant | Cloud Architecture & Automation Special...

Amazing breakdown

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Sonali Kurade Appreciate the feedback.

Like | Reply

 **Onifade-Esan Modupeola Bolaji**  · 3rd+ 3d ...
Founder, Internflare | Growth and Product Marketer | I build audiences and ...

The gap between where teams think they are and where they actually are is real. Most problems I see come from trying to jump layers instead of earning them.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Onifade-Esan Modupeola Bolaji True. The gap between a demo and a deployment is usually foundational debt.

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 **Mohamed Fazil Habeeth** · 3rd+ 2d ...
AI Education & Talent Development | Building Practical AI Capabilities for In...

Clear and well structured breakdown of building AI agents step by step. A great reminder that strong basics are key before moving to advanced systems.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Mohamed Fazil Habeeth Agreed. You can't skip the foundation if you want a system that actually scales.

Like | Reply

 **Alex Ostrovskyy**  · 3rd+ 2d ...
Enterprise AI&MLOps Architect | 9 Years of Hands-on AI on 18-Year Softwar...

Spot on, **Sivasankar**. The chasm between Layer 4 (RAG) and Layer 7 (Function Calling) is where I see most enterprise POCs quietly die. It's one thing to have an LLM hallucinate a summary, but it's a whole different risk profile when you give it API write-access to core systems without rock-solid guardrails.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Alex Ostrovskyy Exactly. The risk profile shifts from "bad text" to "bad data" once APIs are involved..

Like | Reply

 **Michael Lee**  · 3rd+ 2d ...
CRO | Data & AI | Scaling \$1M-\$100M B2B Companies With AI | Turning Lea...

The leap from chatbot to agent is defined by function calling and context management. Without state and memory, your system is architecturally

incapable of handling complex enterprise workflows. Build the reasoning layer before the orchestration layer to avoid compounding hallucinations at scale.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Michael Lee True. Most skip the reasoning layer and wonder why agents fail.

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 **Amrutha Satishkumar**  · 3rd+ 2d ...
Data & AI Solution Engineer @Microsoft | Sharing how Data Platforms shap...

Sivasankar Natarajan The staircase metaphor is accurate. Each higher layer quietly assumes reliability, governance, and clear decision boundaries below it. Skipping steps is where pilots stall.

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 **Sivasankar Natarajan**  Author 2d ...
Technical Director | GenAI Practitioner | Azure Cloud Architect | Data ...

Amrutha Satishkumar Agreed. Reliability is built layer by layer from the bottom up.

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